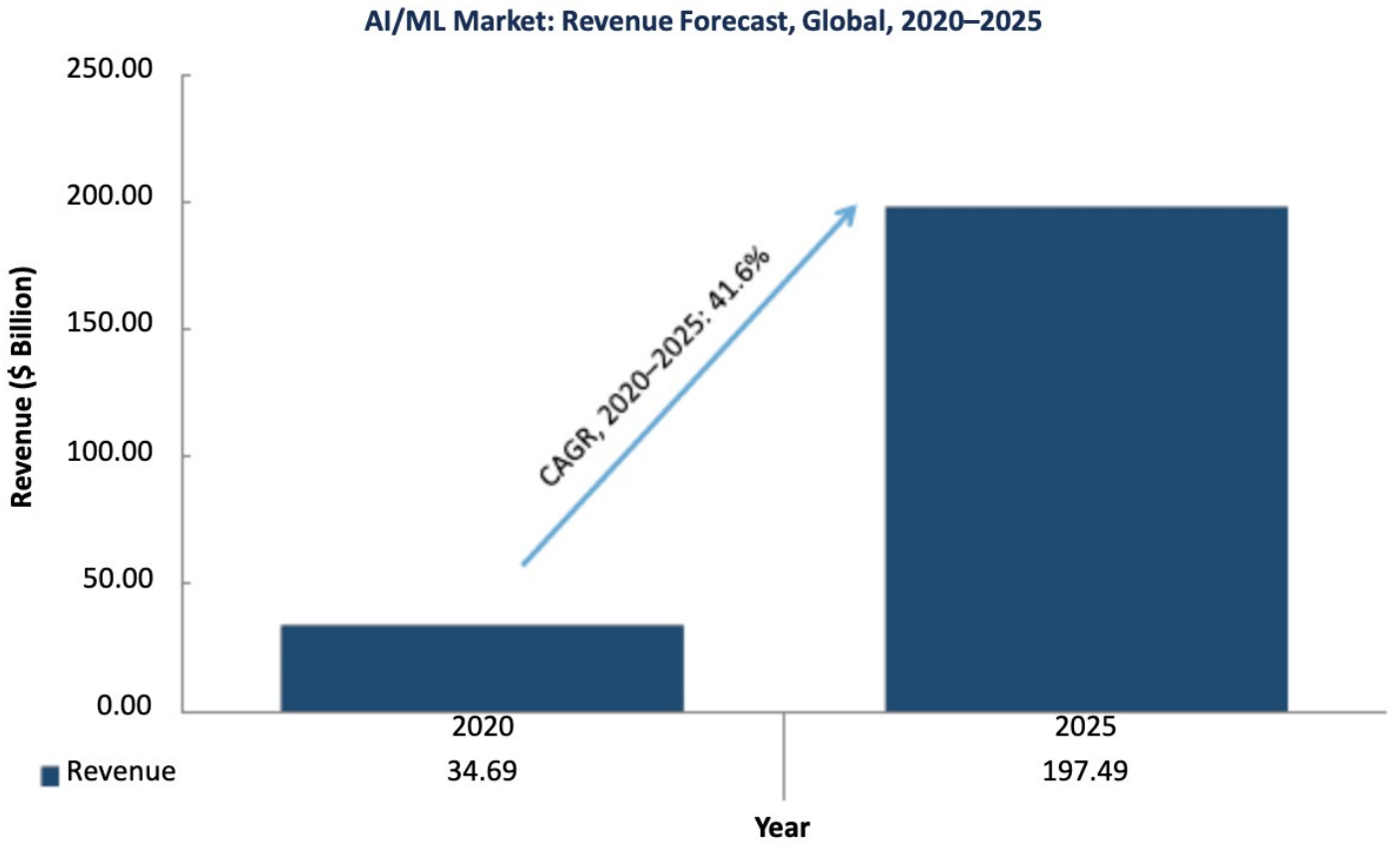
Ambient Systems

AI solutions for Cities of Tomorrow
AI-driven Building Optimization
AI-driven Smart Cities

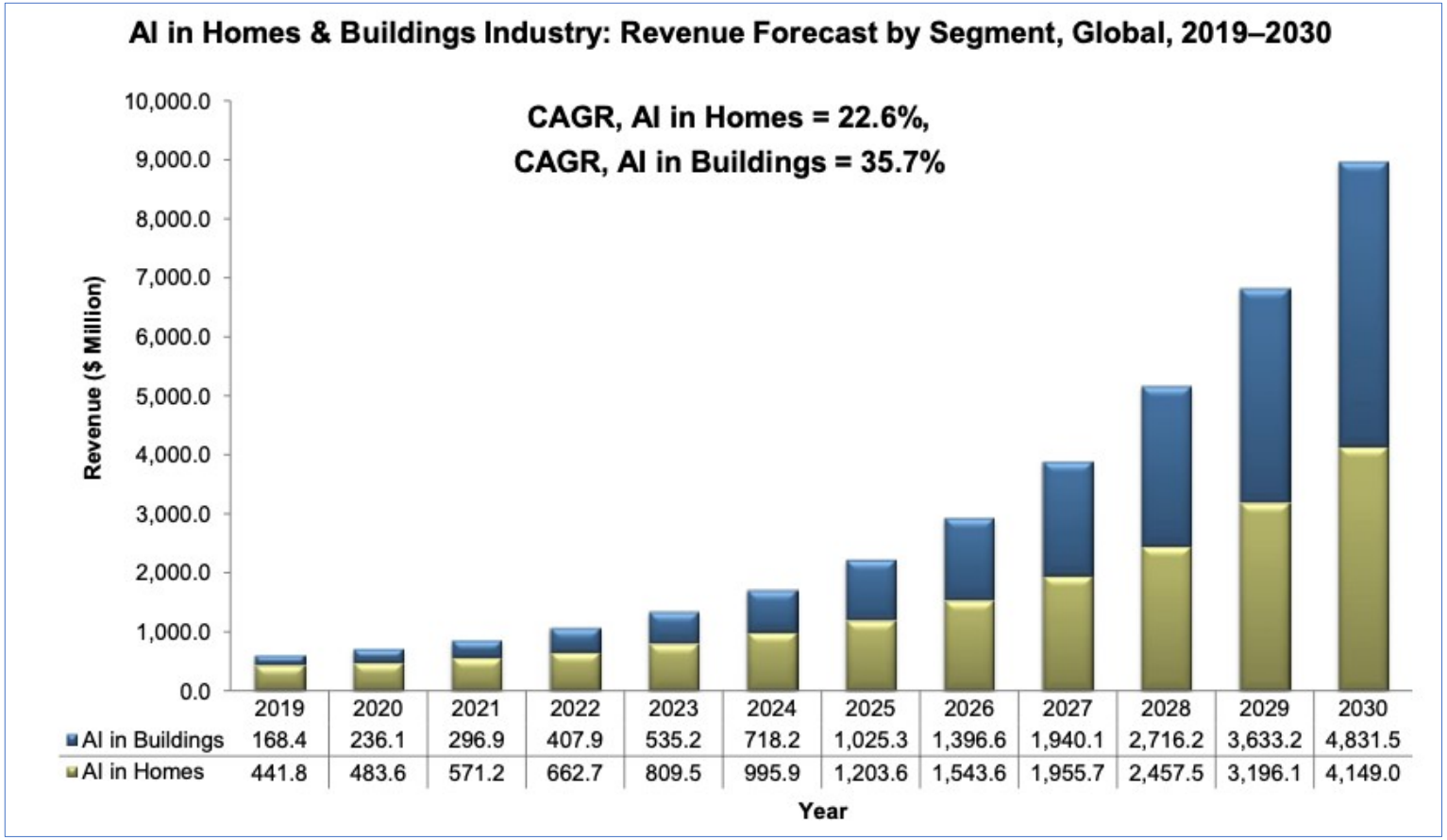
Business Plan Slides

AI/ML Market: $35B growing at 42% CAGR

Revenue Forecast—AI/ML Market



AI Solutions for Smart Buildings



2021 Market Size: $570M
2026 Market Size: $1,400M
CAGR: 36%

25 to 30 companies in the market
Top 5 participants hold 66.3%

Top AI Applications

- Predictive Maintenance
- Building
- Building Process Optimization
- AI-based Optimization
- AI-based Power Trading
- AI-based Lighting Control

Differentiation: Drivers

Driver	Importance	Others	Ambient
Demand for value-driven solutions	H	Mostly focused on energy reduction	Balances competing priorities, such as energy reduction with indoor air quality and productivity
Cost Reduction due to	H	Considers only energy costs	Customers can expect savings by 25% to 35% in terms of not just utilities but also man hours
Increasing Penetration of Transformative Technologies in Buildings	H	Closed hardware or no support for new technologies	Open APIs while consulting the customer on other tech components, including sensors, BMS, and equipment
Government Push for AI Solutions	H	?	Aligned with Singapore government deep tech initiatives
Increasing Awareness on AI-driven Solutions	M	?	Provides online content and thought leadership on the AI-driven solutions

Differentiation: Restraints

Restraint	Importance	Others	Ambient
High CAPEX	H		Customers can start w existing hardware Sensors are optional (if necessary) Modular AI development Fast & simple deployment
Uncertainty of ROI	H		
Cybersecurity	H	?	Add security in design stage Secure default settings Authorization controls Firewalls Stronger end-to-end encryption Threat prevention monitoring
Poor Data Quality	M	?	Efficient data acquisition
Lack of interoperability	L	Closed systems	BACnet compliant Open APIs

AI–Timeline of Adoption

AI, with its machine learning and deep learning features, is key for transforming smart buildings to cognitive buildings.

AI in Homes & Buildings Industry: Timeline of AI Adoption, Global, 2019

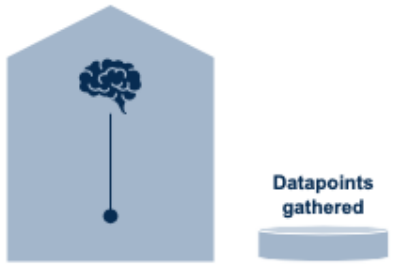


<1980 >2019

Evolution of Building Management
Cognitive buildings to autonomously learn systems and user behaviour to optimise performance

Man-Machine Intelligence Continuum

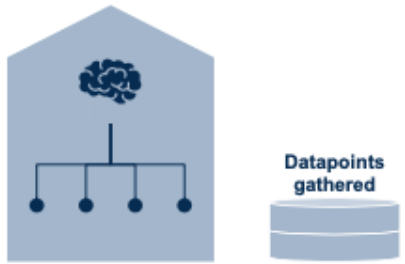
Automated Buildings (1980–2000)



Visualise KPI
- Identify general issues

Assisted Intelligence
- Nature of tasks does not change
- Tasks are automated
- Humans do not learn
- Machines learn
- Examples: Machinery, processes in factories, boilers, ovens

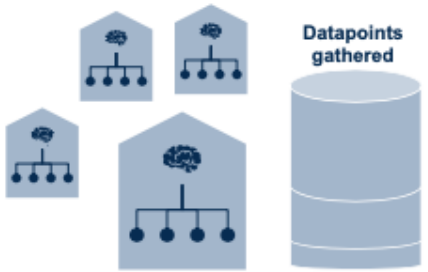
Smart Buildings (2000–2019)



Analyse Energy Consumers
- Understand energy consumption by analysing primary datapoints

Augmented Intelligence
- Nature of tasks changes
- Humans inform machines
- Machines inform humans
- Examples: Predictive maintenance and building performance optimisation

Cognitive Buildings (>2019)

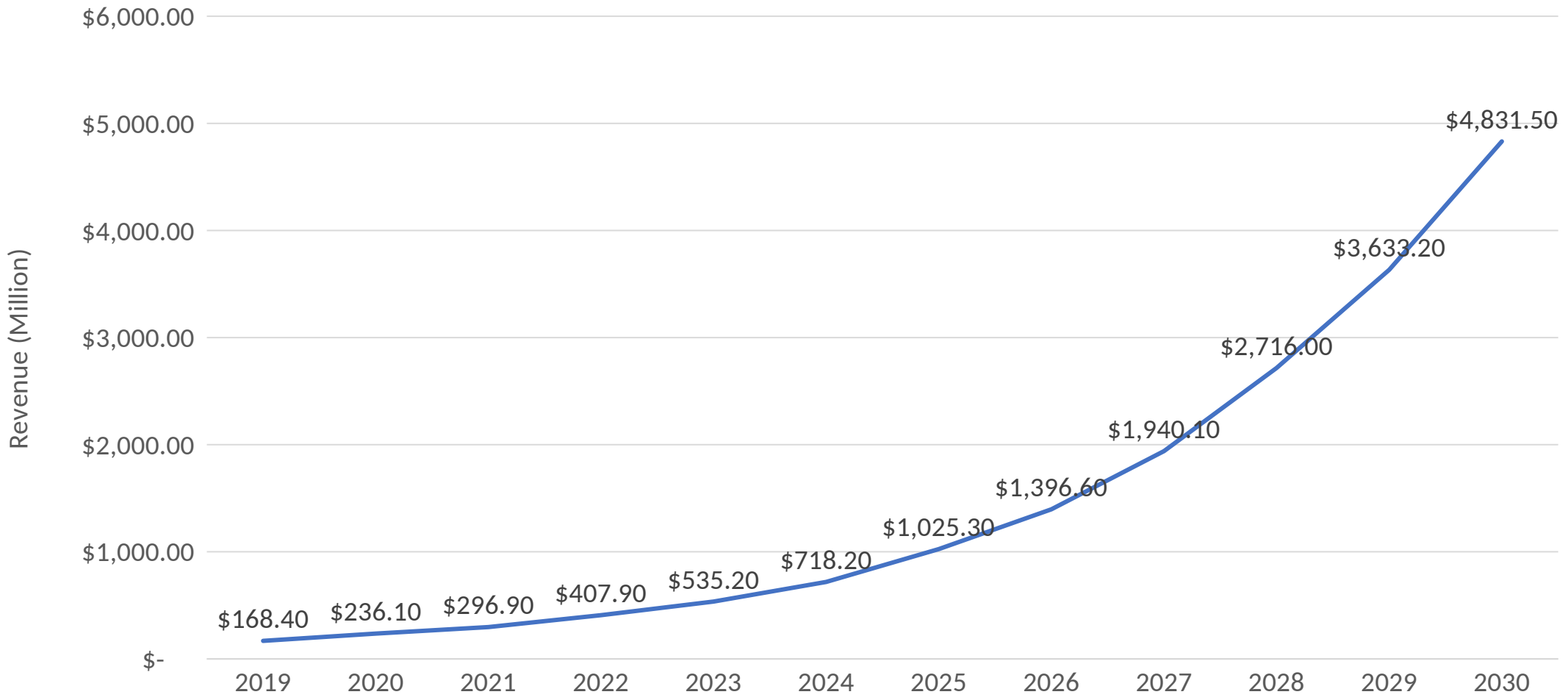


Learn Behaviour
- Personalised control at occupant's level
- Understand energy flow and building occupancy
- Consider occupant comfort
- Collect external datapoints

Autonomous Intelligence
- Nature of tasks changes
- Decisions are automated
- Machines learn continuously
- Example: Digital twin-enabled smart building platform

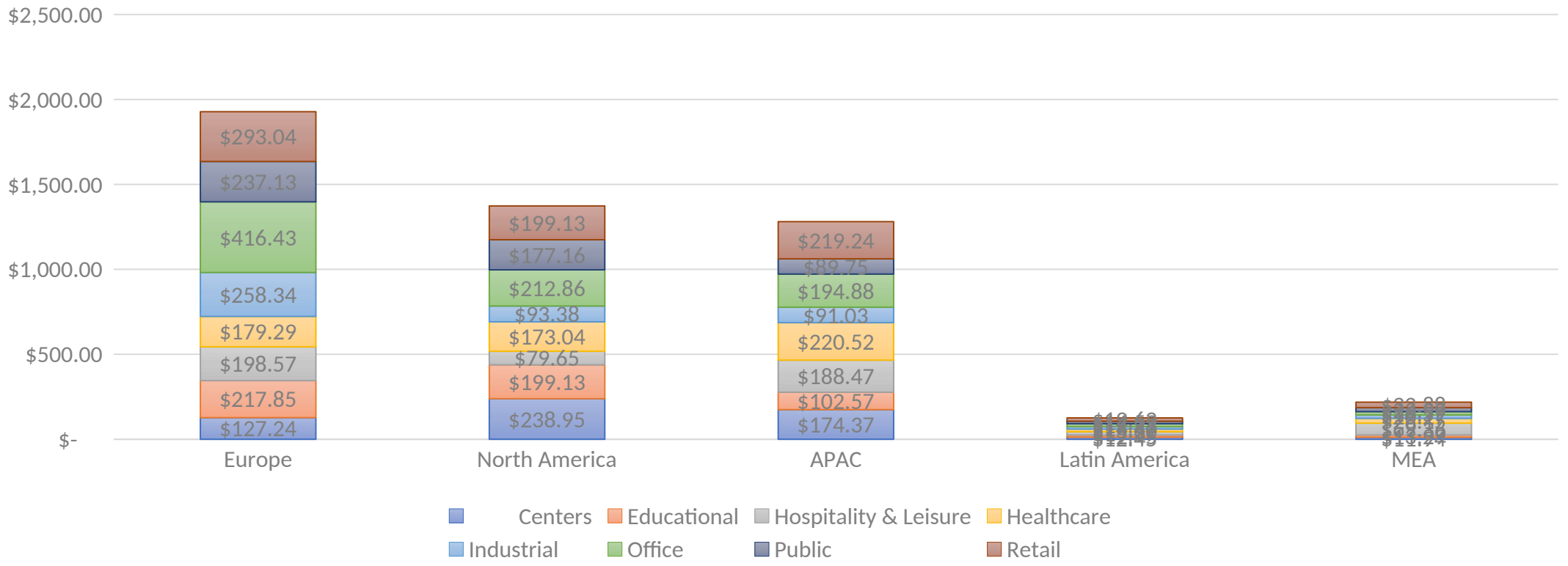
Source: IBM; Frost & Sullivan

AI adoption in buildings is growing fast



Frost & Sullivan

Building Automation Systems Market



Frost & Sullivan

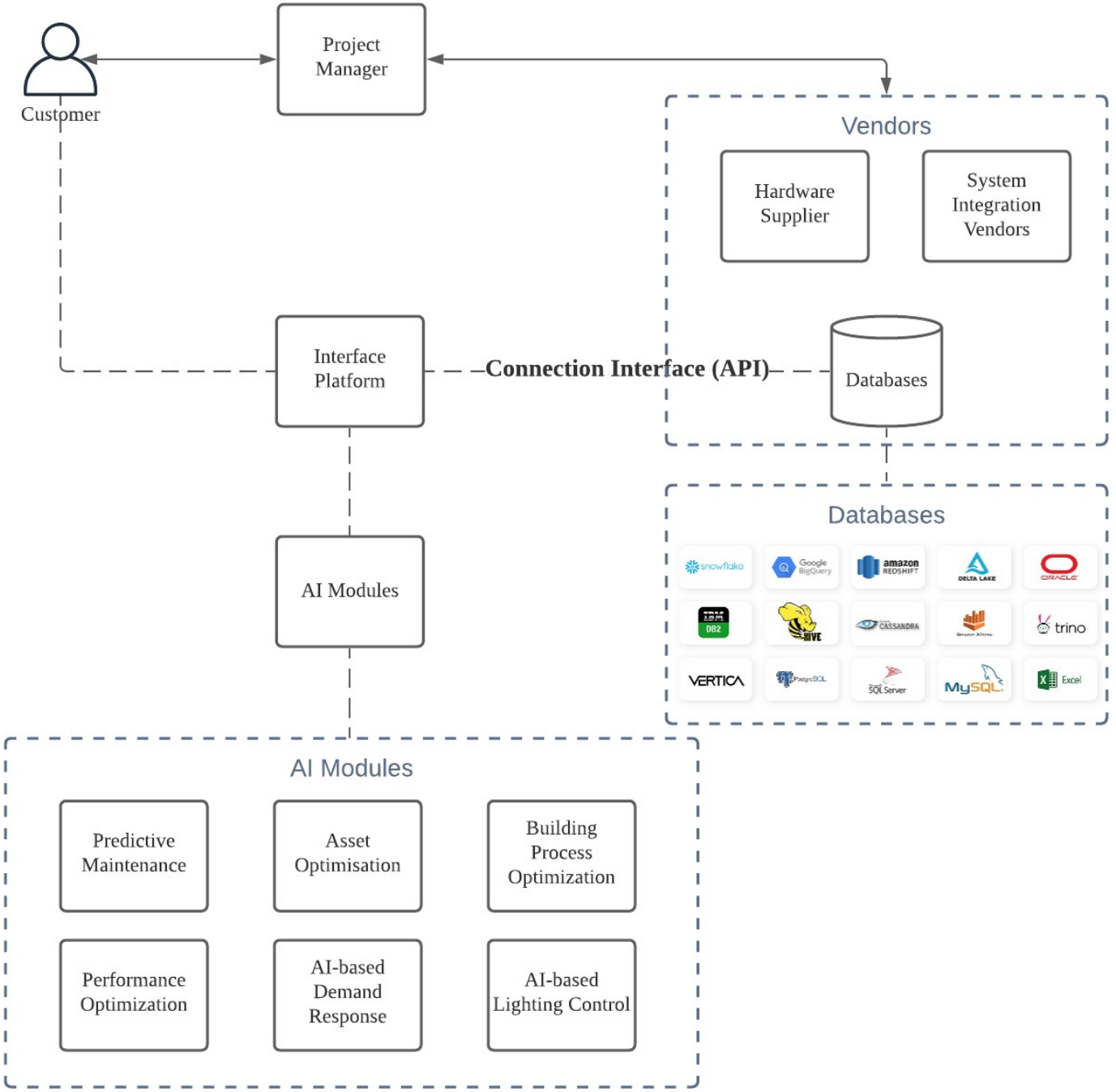
s Offering:

	asset lifetime • Lower cost • Reduced downtime	M to H	H
Building Performance Optimisation	• Increased energy efficiency and reduced cost • Reduced • Greater comfort for occupants	M to H	M to H
Building Process Optimization	• Improved operational efficiency • Reduced response time during critical issues	L to M	M to H
Asset Optimization	• Improved asset lifetime • Real-time optimization of assets • Lower cost • Reduced downtime	M	H
AI-based Demand Response	• Lower energy bills • Production schedule adjustment • Optimized assets	M to H	M
AI-based Power Trading	• Flexibility in power trading • Increased power trading profits	L to M	M
AI-based Lighting Control	• Increased energy efficiency and cost reduction • Enhanced comfort for occupants • Reduced commissioning and configuration time	M	M

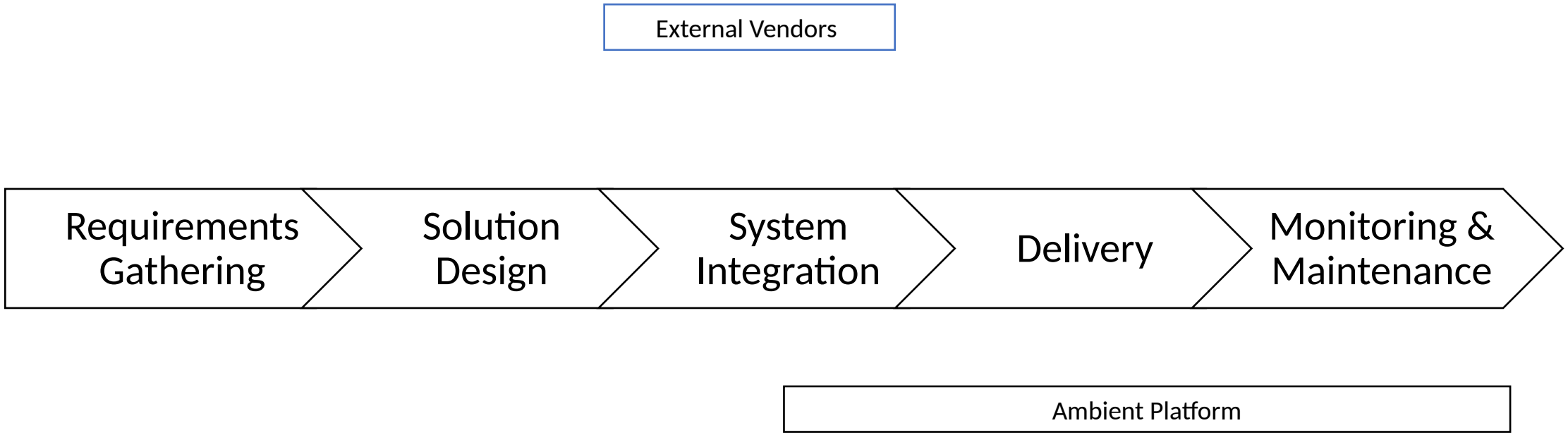
Top Alternatives: Asia and Global

Company	Product	Link
Siemens	MindSphere	https://siemens.mindsphere.io/en
Honeywell	Forge	https://www.honeywell.com/us/en/honeywell-forge/buildings
	Metasys	https://www.johnsoncontrols.com/building-automation-and-controls/building-management/building-automation-systems-bas
Schneider	EcoStruxure	https://www.se.com/sg/en/work/campaign/innovation/overview.jsp
Bosch	Connected Buildings	https://blog.bosch-si.com/projects/bosch-singapore-campus-smart-building-concept-turned-reality/
EcoEnergy Insights	Cortix AI	https://www.cortix.ai/ https://ecoenergyinsights.com/how-we-do-it/proactive-solutions.php#how_we_do_it_video
75F	Facilisight	https://www.75f.io/solutions/products/multi-site-building-automation-75f-facilisight
BrainboxAI	BrainboxAI Dashboard	https://www.brainboxai.com/technology/
BuildingIQ	5i Intelligent Energy Platform	https://buildingiq.com/products/5i-intelligent-energy-platform/
Acuity Brands	Atrius	https://www.acuitybrands.com/brands/internet-of-things/atrius https://www.distech-controls.com/ https://www.acuitybrands.com/products/building-management/unified-building-control

Value Chain


Customer
Project Manager
Vendors
Hardware Supplier
System Integration Vendors
Interface Platform
Connection Interface (API)
Databases
AI Modules
Databases
AI Modules
Predictive Maintenance
Asset Optimisation
Building Process Optimization
Performance Optimization
AI-based Demand Response
AI-based Lighting Control

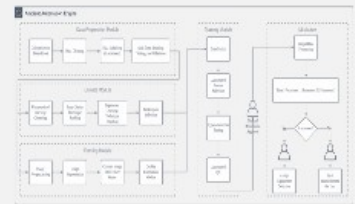
Software + Service Model







Sense



Optimize



Control

SaaS Pricing Model

Service	Item Cost	Total Cost
(SaaS App)		
Page 1: Overview	$500	$3,000
Page 2: Yield Prediction	$1,000	
Page 3: Facility Analytics	-	
Page 4: Maintenance	$1,500	
…		
Data Feeds (API Integrations)		
Data Feed 1: Sensors	$5,000	$10,000
Data Feed 2: System 1	$5,000	
…		
AI Module		
Predictive Maintenance	-	$170,000
Asset Optimization	-	
Building Process Optimization	$50,000	
AI-based Demand Response	$120,000	
AI-based Lighting Control	-	
	-	
	Total Annual Price	$183,000

Professional Services: Project Management

- Priced hourly
- Sample Activities:
  - Problem definition
  - Requirements gathering
  - RFQ preparation
  - Vendor evaluation & selection
  - Vendor management for system integration

Value-based Pricing

AI Module	Value Driver	Annual Value to Customer
Predictive Maintenance	• Improved asset lifetime • Lower cost • Reduced downtime	$100k - $500k
Asset Optimization	• Improved asset lifetime • Real-time optimization of assets • Lower cost • Reduced downtime	$200k+
Building Process Optimization	• Improved operational efficiency • Reduced response time during critical issues	$500k+
	• Increased energy efficiency and reduced cost • Reduced • Greater comfort for occupants	$500k+
AI-based Demand Response	• Lower energy bills • Production schedule adjustment • Optimized assets	$500k+
AI-based Lighting Control	• Increased energy efficiency and cost reduction • Enhanced comfort for occupants • Reduced commissioning and configuration time	$500k+

Annual value per customer based on average facility size of

Growth Strategy

Product	Description	Fee	Growth Targets
A	Platform w Data Feeds	$10,000 per facility	1 per month
B	AI Modules	$100,000 per model	1 per quarter
C	Professional Service	$100 / hr	100 billable hours per month

Ambient Company KPIs

KPI	Current	2021 Goal	2022 Goal
Number of customers	3		
Team Size	5		
Funds Raised	$100k	$500k	$3M
SaaS Subscriptions Sold	0	3	
AI Models Deployed	0	1	5
Professional Service Hours Sold	0	100	
ARR	0		
Area served (sq ft)	0		
$ Savings (energy, carbon footprint, manhours)	$0		
$ Generated	$0		

Stream	V01 Q1 2021: Jan - Mar	V02 Q2 2021: Apr - Jun	V03 Q3 2021:	Full Product Launch Q4 2021: Oct - Dec
Hardware	**Sensors** Collect essential parameters: Temperature, RH, CO2, and light intensity	**Sensors Extended** Collect additional parameters and increase number of sensors	**Collect crop data** Provide hardware to collect crop related data in real-time	Complete Sensors Controls
Dashboard	**On-Site Monitoring** Displays sensor readings in real-time	**Online Monitoring** Real-time visualization, analytics, and notifications	**Notifications** Enhanced visualization and notifications	Interactive and customized
Machine Learning	**Locally Deployed** Combination of real and simulated data	**Deploy on the cloud** Train using additional data and deploy through the online dashboard	**Fully Trained Model** Distribute training at the edge and in the cloud for real-time insights	Trained and deployed in real-time

Operations Plan – Year 1 and 2 (2021-2022)

	2021				2022			
	Q1 2021	Q2 2021	Q3 2021	Q4 2021	Q1 2022	Q2 2022	Q3 2022	Q4 2022
	Jan – Mar	Apr – Jun	Jul – Sep	Oct - Dec	Jan – Mar	Apr – Jun	Jul – Sep	Oct - Dec
Customer pipeline	2	3	5	10	15	20	25	30
Team Size	2	3	6	6	6	8	8	10
ARR: software	-	-	$10k	$30k	$30k	$30k	$30k	$30k
ARR: AI	-	-	-	$100k	$100k	$100k	$100k	$100k
Professional Services	-	-	$30k	$30k	$30k	$30k	$30k	$30k
Funding cycle	-	-	Pre-seed ($50k)	Pre-seed ($150k)		Seed ($2M)		
Markets		Singapore				Singapore		
	-	MVP for customer development	Live Software + AI Demo	Production Software + AI		New AI modules		

Operations Plan – Year 2 and 3 (2023-2024)

	2023				2024			
	Q1 2023	Q2 2023	Q3 2023	Q4 2023	Q1 2024	Q2 2024	Q3 2024	Q4 2024
	Jan – Mar	Apr – Jun	Jul – Sep	Oct - Dec	Jan – Mar	Apr – Jun	Jul – Sep	Oct - Dec
Customer pipeline	~30 to 50				~ 50 to 70			
Team Size	10 to 15				15 to 25			
ARR: software								
ARR: AI								
Professional Service								
Funding cycle				Series A ($10M)				
Markets	Singapore		Singapore Hong Kong		Singapore Hong Kong		Singapore Hong Kong Japan	
		New AI modules				New AI modules		

Operations Plan – Year 4 and 5 (2025-2026)

	2025				2026			
	Q1 2025	Q2 2025	Q3 2025	Q4 2025	Q1 2026	Q2 2026	Q3 2026	Q4 2026
	Jan – Mar	Apr – Jun	Jul – Sep	Oct - Dec	Jan – Mar	Apr – Jun	Jul – Sep	Oct - Dec
Customer pipeline	~70 to 90				~100			
	to 50							
ARR: software								
ARR: AI								
Professional Service								
Funding cycle								
Markets	Singapore Hong Kong Japan				Singapore Hong Kong Japan			
	New AI modules				New AI modules			

Appendix

Competitors

Mapping of Companies Offering AI Applications in the H&B Industry

Key participants offering AI applications.



Image source: Freepik.com

Source: Frost & Sullivan

Top AI Companies to Watch

Leading participants of building technologies.

Company	Growth	Innovation
SIEMENS	Siemens Smart Infrastructure drives the digital transformation of the buildings industry by creating open-standards-based Building Information Modelling (BIM)-ready products and services and by connecting its products and installed base to the new digital offerings of Siemens Comfy and Enlighted in the smart space area. Siemens' global team of data scientists are involved in the development of a next-generation AI platform for data-driven applications as a part of the company's digital services.	Siemens' MindSphere platform collects data from numerous factory and industry equipment and provides digital solutions such as remote diagnostic services, performance optimiser, remote maintenance, and digital whiteboard. The company might leverage this experience and develop a new data-driven application for smart buildings in collaboration with IBM.
Honeywell	Honeywell's building technologies (HBT) business in the non-residential segment is expected to continue its strong performance in the commercial fire safety domain and record modest growth in security solutions. Honeywell, after the spin-off of its homes business, Resideo, is well-positioned to grow profitably, with a 23% margin target by 2023.	Honeywell has announced the launch of Honeywell Forge Energy Optimization, a cloud-based, closed-loop, machine learning solution that studies the pattern of building energy consumption and continuously optimises building assets for improved energy efficiency without compromising on occupant's comfort.
Signify	Signify's light-emitting diode (LED)-based sales reached 78% of its total revenue, growing at a rate of 2.7% in 2019. LED-based sales are estimated to account for 85% of Signify's total sales in 2020, and its connected light points installed base is expected to exceed 65 million in 2020.	Signify will continue to strengthen its growth in data-enabled services by delivering IoT-based lighting solutions. Indoor and outdoor segment customers will benefit from the Interact IoT platform that leverages AI to deliver cost and operational efficiencies.
AcuityBrands.	Acuity Brands is the leading participant in the LED and non-LED luminaire market in the Americas, with 85% of its sales coming from the non-residential segment and 15% from the residential segment.	Acuity Brands' integrated package includes the Atrius sensory network, the building OS platform, and the Atrius platform that has been deployed for 320 million sq,ft. of space.

Source: Frost & Sullivan

Top AI Companies to Watch (continued)

High-performing and emerging companies.

Company	Growth	Innovation
ECOENERGY INSIGHTS	For EcoEnergy Insights, 2019 was an exceptional year in terms of both organic growth and product innovation. Frost & Sullivan recognised the company's efforts and awarded it for demonstrating growth, innovation, and leadership in the global BEMS market. The company is expected to continue its strong performance in 2020.	EcoEnergy Insights' CORTIX platform has been upgraded to serve 19 categories of building equipment in different end-user verticals. Overall, 150,000 equipment are connected to the CORTIX platform. The platform's base engine is AI- and ML-driven, providing Insights-to-Action and, thereby, creating value for customers. More updates are expected in 2020.
FOGHORN	FogHorn ensures that its current product line and future road map remain relevant to its customers. The company focusses on improving the performance value for clients through an efficiency value play strategy. FogHorn has a dedicated global customer service team, with strong presence in Japan, Korea, Saudi Arabia, and the United States.	FogHorn leverages its industry-leading edge analytics capabilities by introducing features to its smart building AI solutions that include air quality monitoring, facial recognition with deep learning, video-based security surveillance for increased comfort and safety, and early water pump failure detection through sensory functionality.
75F	75F raised $18 million in Series A funding in 2019, led by Breakthrough Energy Ventures and Oil & Gas Climate Initiative. This funding will help the company invest in research and development (R&D) for product innovation and strengthening its partner network in North America, Middle East, and Asia-Pacific (APAC). The company expects to register more than 100% growth in 2020.	75F's self-learning building intelligence system improves a commercial building's operational efficiency and occupant experience by proactively controlling the HVAC, indoor air quality, and lighting settings based on inputs from sensors, weather data, and configuration settings and predictive algorithms powered by machine-to-machine (M2M) engine and cloud-computing data storage and processing power.

Source: Frost & Sullivan

Top AI Companies to Watch (continued)

High-performing and emerging companies.

Company	Growth	Innovation
BRAINBOX AI	BrainBox AI has been growing at lightening speed. The company, which was working with less than half a million square feet in early 2019, hit 10 million by the end of the year. Currently, it manages around 18 million sq.ft. of space including retail spaces, offices, hotels, schools, warehouses, datacentres, hospitals, and grocery stores.	BrainBox AI's Artificial Intelligence engine uses advanced predictive control strategies with ML techniques that can self-learn a building's operating behaviour and energy flows and analyse and instruct the HVAC system to operate more efficiently, thereby reducing operating costs and carbon footprints significantly.
uplight	Tendril and Simple Energy merged as Uplight, with 75 North American utilities as customers. Earlier, Tendril had been on an acquisition spree, taking over EEme, an energy disaggregation start-up; EnergySavvy, a residential customer engagement software company; and FirstFuel Software, an energy efficiency data analytics company.	Uplight's combined business capabilities is a three-step process: engage, activate, and orchestrate. The first step is covered by Tendril's customer home energy reports and online interactions; the second step is covered by Simple Energy's software tools and management platforms to convert passive customers to active customers; and the third step is covered by Tendril's data analytics capability.
sense	Sense raised $18 million in fresh funding from investors led by Schneider Electric and including Shell Ventures, iRobot, Energy Impact Partners, Prelude Ventures, and Capricorn Investment Group. The company has partnered with 50 solar installers in more than 24 states in the US and Canadian provinces where home owners are embracing solar power.	Sense uses high-resolution data to detect small changes in magnitude, phase, and frequency and then incorporates machine learning algorithms to distinguish one appliance from another. The Sense Home Energy Monitor will be compatible with Google Assistant and the Belkin Wemo Insight Plug and TP-Link Kasa HS110 smart plugs in 2019.

Source: Frost & Sullivan

Top AI Companies to Watch (continued)

High-performing and emerging companies.

Company	Growth	Innovation
BuildingIQ	BuildingIQ has been recording a CAGR of more than 20% since 2015 in the global BEMS market. Of late, the company has been busy expanding its partner network. BuildingIQ has raised a total of $6.5 million since it announced its capital raising in 2017. The company has an established market presence in the US, especially for large building portfolios.	BuildingIQ provides energy management solutions through its 5i energy intelligence AI platform for a wide range of buildings. Its solutions include a suite of services such as visualisation, building operations, and predictive control. The platform learns and evolves with existing building management systems and customer needs.
PointGrab	The Israeli smart building management company, PointGrab is an emerging firm that has been growing at an accelerated rate over the past few years. The company is backed by Signify, ABB Technology Ventures, and Mitsubishi Capital. It partners with Gooee for smart lighting, Serraview for connected workplace, and Tyco (now part of JCI). PointGrab has access to a large customer base through its partners.	The CogniPoint AI-based smart sensing platform is a combination of occupancy sensors and cloud management system. The occupancy sensors continuously send workplace occupancy data to the cloud management system, which in turn shares the data with a third-party building management system for overall building performance optimisation.
smappee	The Belgian home energy monitoring start-up, Smappe is estimated to have recorded a CAGR of more than 50% since 2015 in the home energy management systems (HEMS) market. The company's products are compatible with those offered by most of the connected home solution providers such as Google, Amazon, Varta, Niko, Netatmo, Daikin, Tado, and Signify. The company's products support IFTTT, Conrad Connect and MQTT, OCPP, Modbus, Bluetooth, KNX and more protocols.	The AI-based home energy monitoring device detects every appliance for energy consumption and generation through the incorporation of machine learning algorithms. The home energy monitoring system consists of hardware devices for tracking electricity, gas, and water consumption. It is installed in the breaker panel or on appliances to track energy consumption and energy anomalies of appliances via a dashboard application.

Source: Frost & Sullivan

Top AI Companies to Watch (continued)

High-performing companies.

Company	Growth	Innovation
CURB	Energy Curb, a US-based home monitoring start-up, has been growing at an accelerated rate over the past couple of years. The company has partnered with Schneider Electric to integrate its home energy monitoring product with Schneider's Square D smart load breaker panel, a modular product. This partnership has helped Schneider Electric enter the home energy management market and allowed Energy Curb to expand its customer base.	The AI-based home energy monitoring device detects every appliance by incorporating machine learning algorithm. The product is a hardware device installed in the breaker panel and a mobile application to track energy consumption and energy anomalies of appliances.
VELUX	The Danish company is one of the leaders in roof windows, skylights, and intelligent home controls. In 2017, the company partnered with Netatmo to offer an indoor climate control solution to its residential customers. Frost & Sullivan considers it as a highly impactful, value-driven, AI-based solution available in the residential market.	Velux ACTIVE is an indoor climate control solution that includes a Wi-Fi gateway and smart controls for Velux INTEGRA roof windows and sunscreening products. The solution incorporates a machine learning algorithm for ventilation and blind control.
VATTENFALL	Vattenfall continues to invest significantly in renewable production. An investment of about $2.46 billion is planned for wind farms, of which 75% is for offshore wind farms. The company is also investing $100 million in solar and battery projects and another $170 million in distribution network solutions and decentralised heat solutions. The company has bought 100% shares of Senfal and is adding new services for industrial demand-side flexibility and renewable generation optimisation.	Vattenfall's acquisition of Senfal has made it one of the very few companies offering autonomous demand response and power trading solutions. Senfal's AI algorithm enables renewable energy operators and large industrials to optimally use their assets, and the platform makes autonomous decisions on energy markets.

Source: Frost & Sullivan